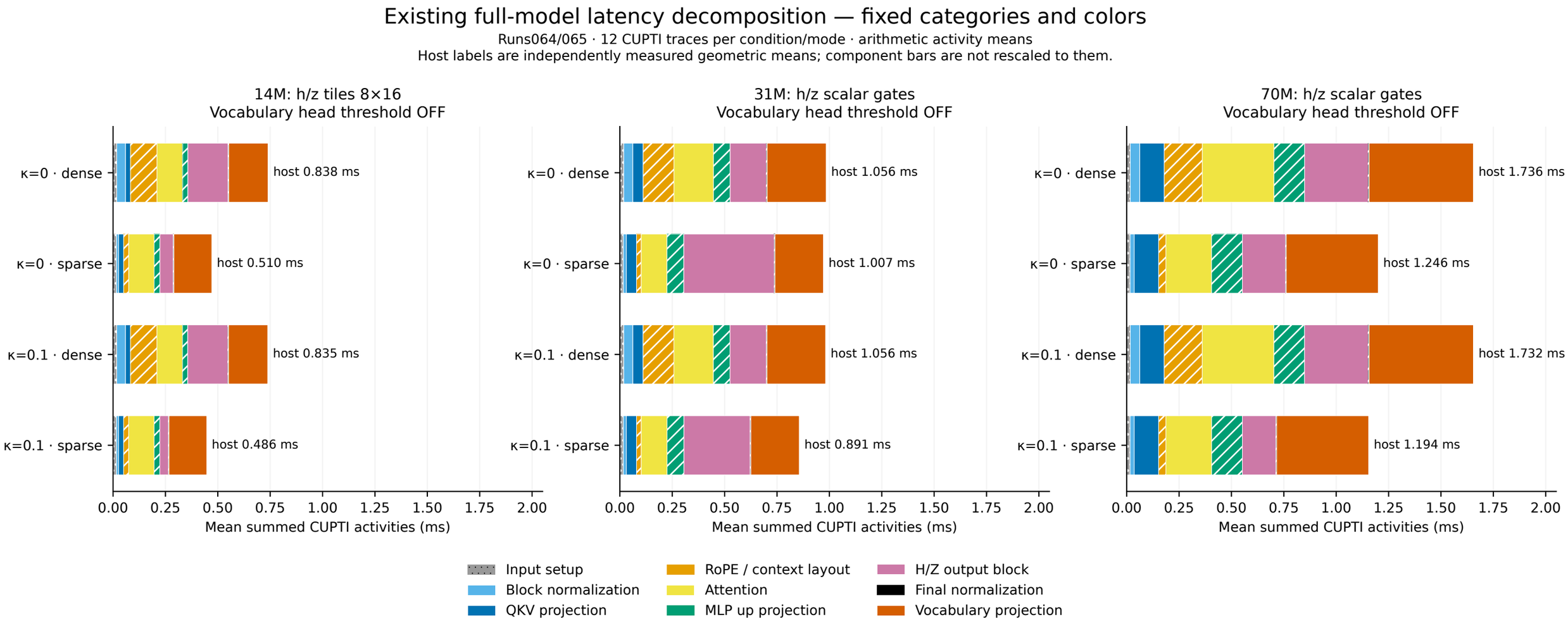

Existing full-model latency decomposition — fixed categories and colors
Runs064/065 · 12 CUPTI traces per condition/mode · arithmetic activity means
Host labels are independently measured geometric means; component bars are not rescaled to them.
14M: h/z tiles 8×16
Vocabulary head threshold OFF
31M: h/z scalar gates
Vocabulary head threshold OFF
70M: h/z scalar gates
Vocabulary head threshold OFF
κ=0 · dense
κ=0 · sparse
κ=0.1 · dense
κ=0.1 · sparse
host 0.838 ms
host 0.510 ms
host 0.835 ms
host 0.486 ms
host 1.056 ms
host 1.007 ms
host 1.056 ms
host 0.891 ms
host 1.736 ms
host 1.246 ms
host 1.732 ms
host 1.194 ms
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Mean summed CUPTI activities (ms)
Input setup
Block normalization
QKV projection
RoPE / context layout
Attention
MLP up projection
H/Z output block
Final normalization
Vocabulary projection